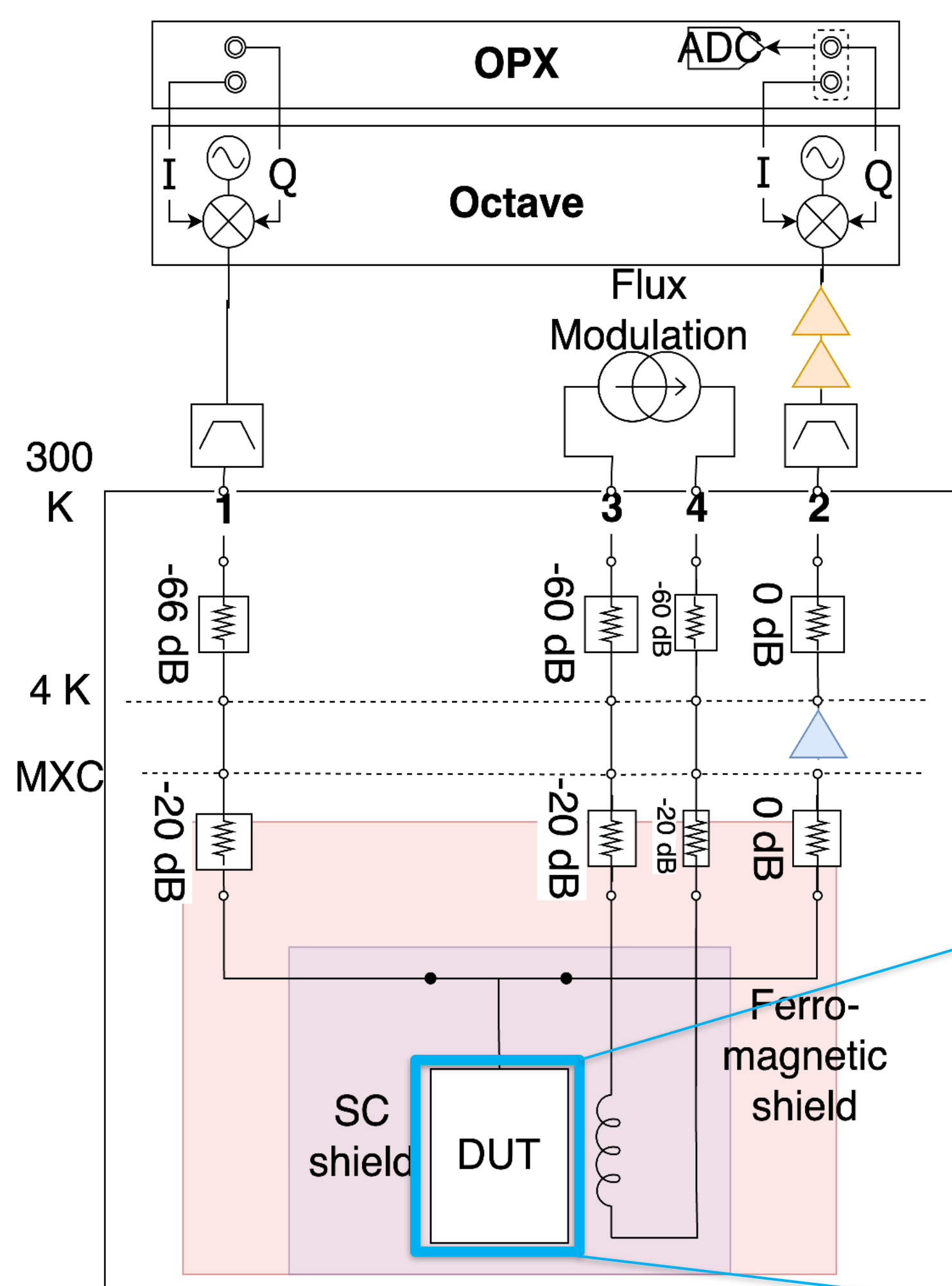


Noise Characterization of Flux Tunable Resonators

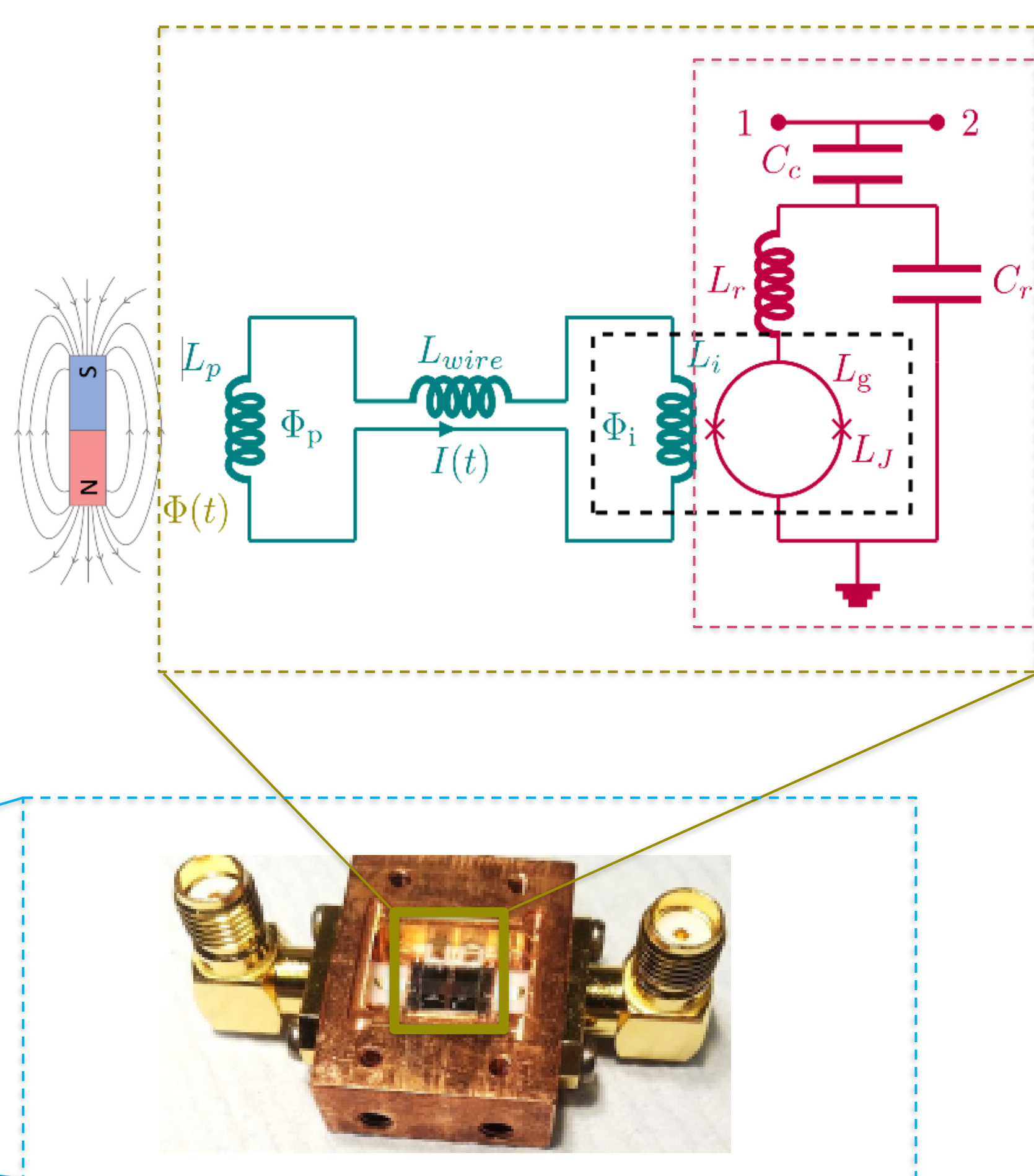
Motivation

A flux-tunable resonator (FTR) is a planar superconducting quarter-wave resonator terminated by a DC-SQUID (Superconducting QUantum Interference Device) at the ground end, which enables the tuning of its resonance frequency by an external magnetic flux. FTRs are used in various applications such as ultra-low field magnetometry or used as a tunable qubit coupler. This flux sensitivity comes with a cost: FTRs are also very sensitive to other flux sources such as stray magnetic fields or noise from the instrument. A flux noise characterization is therefore needed to identify the nature and sources of the noise present, and then further develop techniques to cancel the effects of flux noise.

Measurement schematic



FTR Based Flux Readout

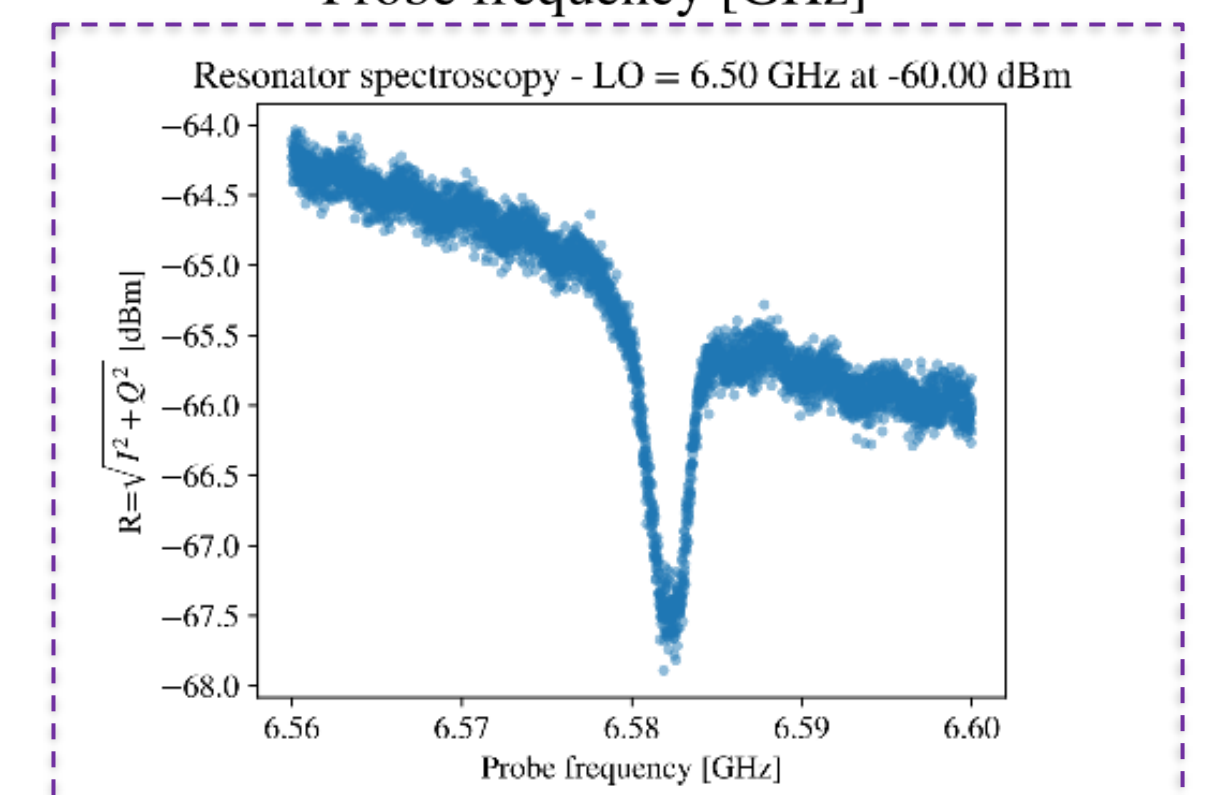
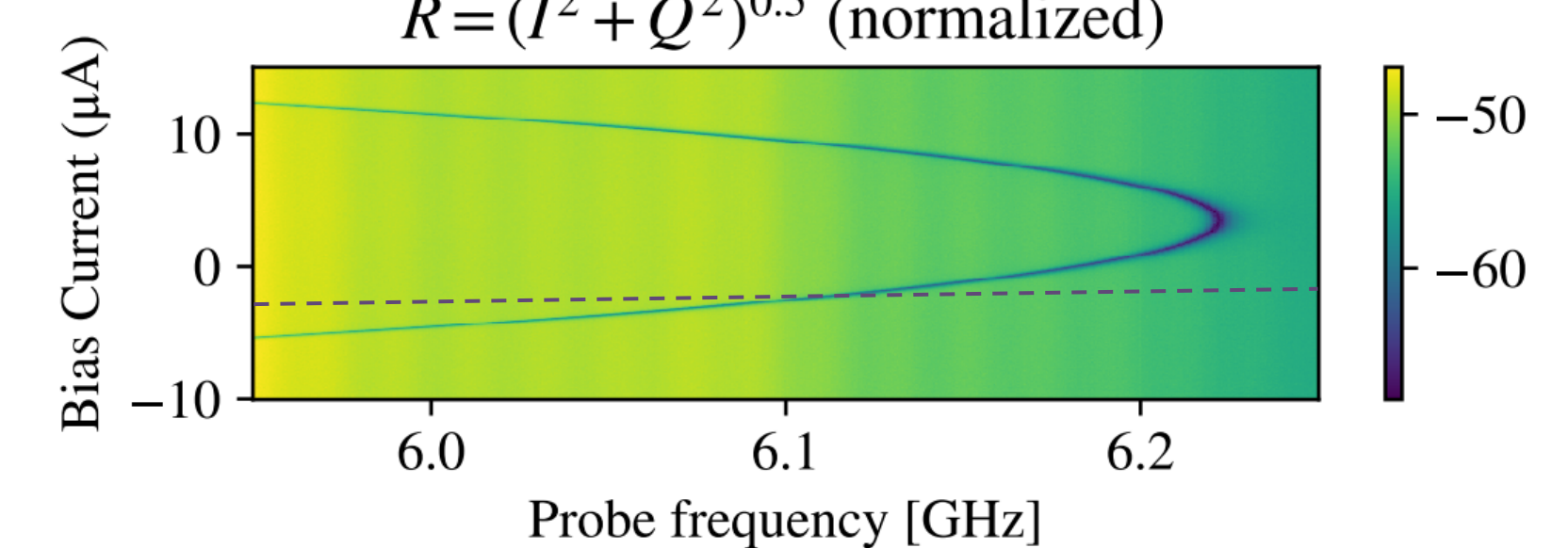


Flux-Tunable Resonator

The resonant frequency of the FTR includes the inductance from Josephson Junction (JJ), which is a function of flux thereby making the FTR tunable.

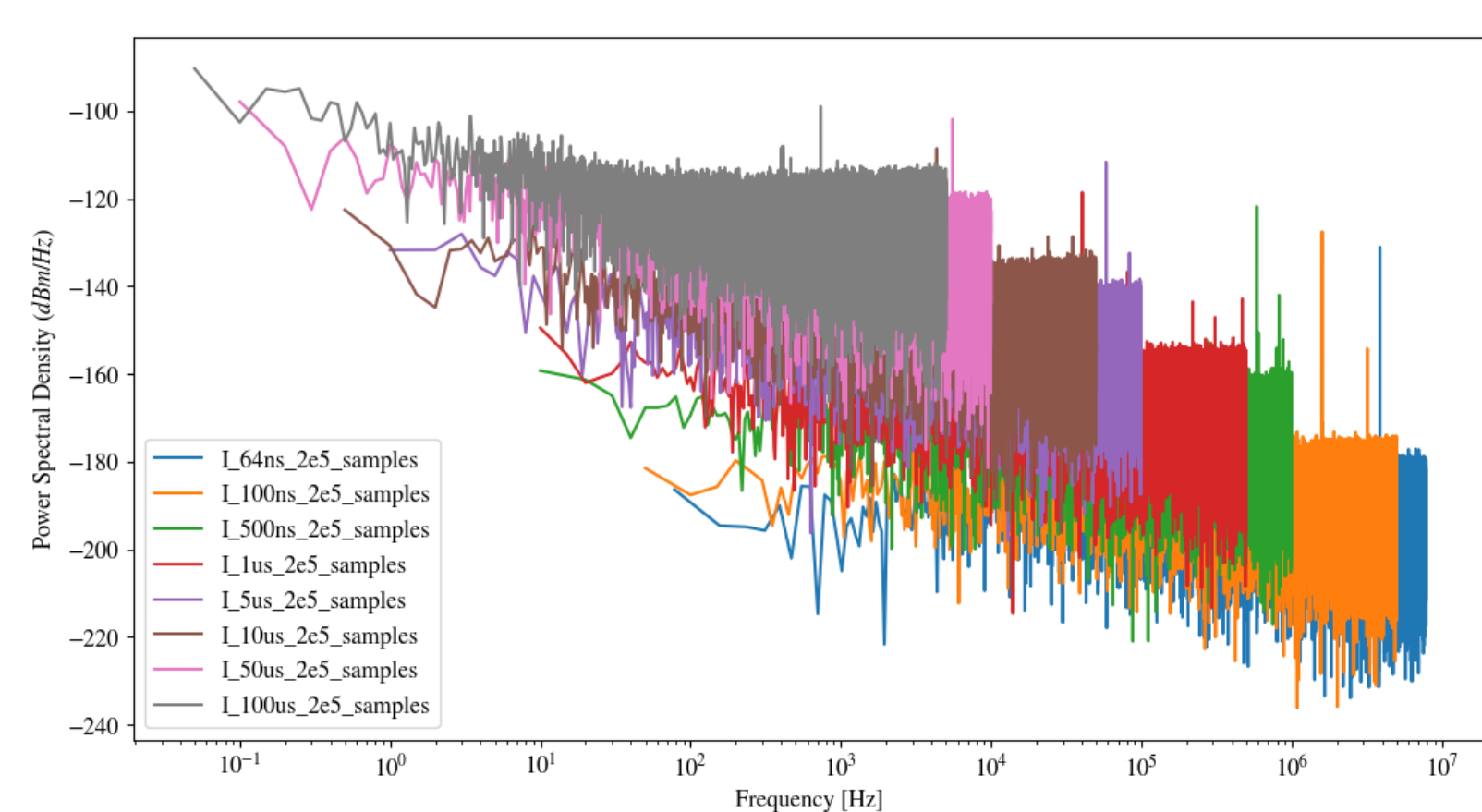
$$\omega_r = \sqrt{\frac{1}{[L_r + L_{geo} + L_j(\phi)/2] \times C_r}}$$

$$R = (I^2 + Q^2)^{0.5} \text{ (normalized)}$$



Noise Characterization

Noise characterization of the FTR is done by measuring the amplitude and phase of the signal from the FTR and calculating the Power Spectral Density



The frequency range of the PSDs depend on the sample size & frequency

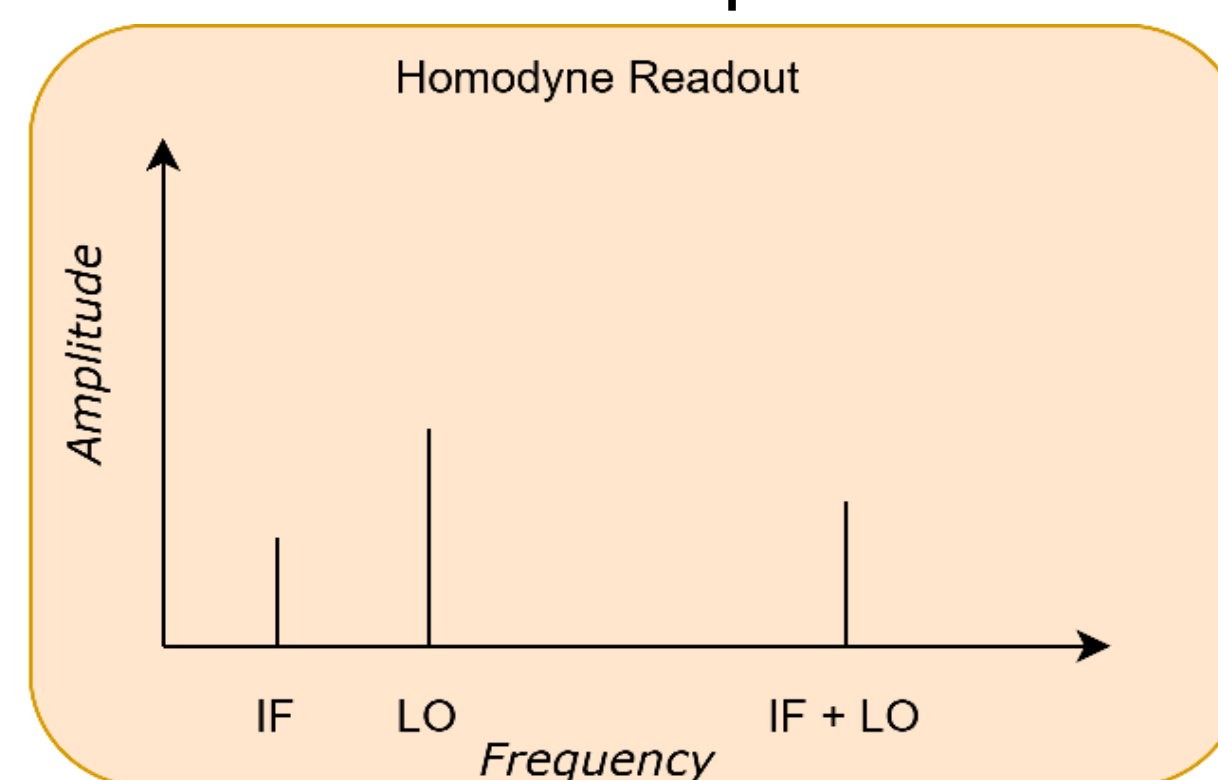
The expected noise sources are:

- Electronic noise**
 - Flicker noise (1/f)
 - Instrument noise
 - Amplifier noise
 - Ground noise.
- Flux noise**
 - Earth's field
 - Stray fields from cable
 - Vibration induced fields
- Parasitic TLS noise**

Methodology

The FTR signal is acquired using an FPGA-based control and measurement system^[2]. We then perform a homodyne readout at FTR resonance^[1].

- LO □ Reference Freq for upconversion
- IF □ Intermediate Freq of the FPGA



Outlook

- Characterize the resonator at different powers and temperatures for on and off resonance points.
- Investigate the sources of noise based on the results obtained
- Develop noise cancellation technique based on Pound-Drever Hall locking [3]

Power Spectral Density

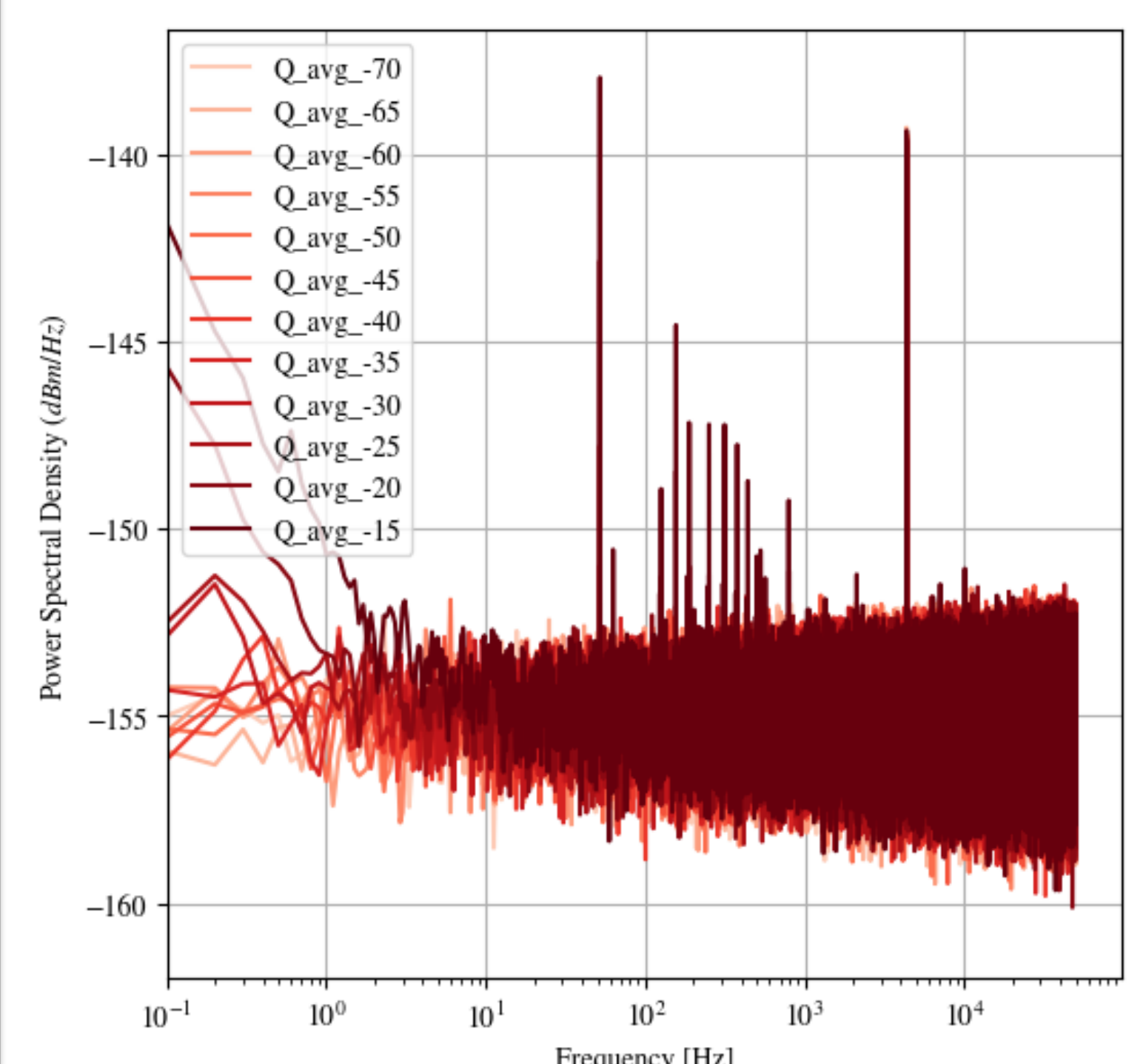
The Power Spectral Density $S_x(f)$ of a signal $x(t)$ is defined as the Fourier transform of its auto correlation function

$$S_x(f) = \int_{-\infty}^{\infty} R_x(\tau) e^{-i2\pi f\tau} d\tau$$

where,

$$R_x(\tau) = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T x(t) * x(t + \tau) dt$$

Preliminary Results are shown below



On resonance Q Plots (Bare Resonator)

LO = 4 GHz

IF = 275.333 MHz

References

- [1] Gao, J., Zmuidzinas, J., Mazin, B. A., LeDuc, H. G. & Day, P. K. Noise properties of superconducting coplanar waveguide microwave resonators. *Applied Physics Letters* **90**, 102507 (2007).
- [2] Hybrid control for quantum computing. *Quantum Machines* (2025). Available at: <https://www.quantum-machines.co/>.
- [3] Drever, R. W. *et al.* Laser phase and frequency stabilization using an optical resonator. *Applied Physics B Photophysics and Laser Chemistry* **31**, 97–105 (1983).